Note 5: QEC1

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5.1 Summary of Progress

1. Studied the general framework of classical error correction models.
2. Reviewed binary linear codes, parity-check matrices and syndrome decoding.
3. Introduced fundamental concepts of quantum error correction, stabilizer and CSS codes.

5.2 Classical errors correction

Definition 1 A **general error-correction model** consists of:
- $\mathcal{M}$: message set,
- $\mathcal{X}$: codeword set,
- $\mathcal{E}$: set of error patterns, and for each $e \in \mathcal{E}$, a channel map $\tau_e\colon \mathcal{X} \to \mathcal{X}$,
- encoding map $\mathrm{Enc}\colon \mathcal{M} \to \mathcal{X}$.
- decoding map $\mathrm{Dec}\colon \mathcal{X} \to \mathcal{M}$.

The model is **error-correcting** if for all $e \in \mathcal{E}$, the diagram commutes:

$$\begin{tikzcd} \mathcal{M} \arrow[r, "\mathrm{Enc}"] \arrow[d, "\mathrm{Id}"'] & \mathcal{X} \arrow[d, "\tau_e"] \\ \mathcal{M} & \mathcal{X} \arrow[l, "\mathrm{Dec}"] \end{tikzcd}$$

Equivalently,

$$\mathrm{Dec} \circ \tau_e \circ \mathrm{Enc} = \mathrm{Id}_{\mathcal{M}}.$$

Lemma 2 *In general error-correction model, the map* Enc *is injective, denote* $\mathcal{C} = \mathrm{Im}\,\mathrm{Enc}$ *, recover map* $\mathcal{R} := \mathrm{Enc} \circ \mathrm{Dec}$*, then we have*

$$\mathcal{R} \circ \tau_e = \mathrm{Id}_{\mathcal{C}}.$$

Proof $\mathrm{Dec} \circ \tau_e \circ \mathrm{Enc} = \mathrm{Id}_{\mathcal{M}} \Rightarrow \mathrm{Dec} \circ \tau_e = \mathrm{Enc}^{-1}|_{\mathcal{C}}$, So

$$\mathcal{R} \circ \tau_e = \mathrm{Enc} \circ \mathrm{Dec} \circ \tau_e = \mathrm{Enc} \circ \mathrm{Enc}^{-1}|_{\mathcal{C}} = \mathrm{Id}_{\mathcal{C}}$$

■

The model can be simplified as

$$\begin{tikzcd} \mathcal{C} \arrow[rr, equal, "\mathrm{Id}"] \arrow[dr, "\tau_e"'] & & \mathcal{C} \\ & \mathcal{X} \arrow[ur, "\mathcal{R}"'] & \end{tikzcd}$$

Definition 3 A subset $\mathcal{C} \subseteq \mathbb{F}_2^n$ is an $[n,k,d]$-Binary Linear code if

$$\mathcal{C} \leq \mathbb{F}_2^n, \quad \dim_{\mathbb{F}_2} \mathcal{C} = k, \quad d = \min_{\boldsymbol{c} \neq \boldsymbol{c}' \in \mathcal{C}} d_H(\boldsymbol{c}, \boldsymbol{c}').$$

Definition 4 Let $\mathcal{C} \leq \mathbb{F}_2^n$ be an $[n,k,d]$ linear code.

1. An $(n-k) \times n$ matrix $H \in \mathbb{F}_2^{(n-k)\times n}$ is called the **parity-check matrix** of $\mathcal{C}$, if

$$\mathcal{C} = \left\{ \boldsymbol{c} \in \mathbb{F}_2^n \,\middle|\, H\boldsymbol{c}^\top = \boldsymbol{0} \right\}.$$

2. For any received vector $\boldsymbol{r} \in \mathbb{F}_2^n$, the vector

$$\boldsymbol{s} = H\boldsymbol{r}^\top \in \mathbb{F}_2^{n-k}$$

is called the **syndrome** of $\boldsymbol{r}$.

Remark 5 The parity-check matrix is not uniqueness.

Theorem 6 *Let $\mathcal{C} \subseteq \mathbb{F}_2^n$ be an $[n,k,d]$ linear code. There exists a basis $\{\boldsymbol{g}_1, \boldsymbol{g}_2, \ldots, \boldsymbol{g}_k\}$ of $\mathcal{C}$ such that the generator matrix formed by these basis vectors can be written as*

$$G = \begin{pmatrix} \boldsymbol{g}_1 \\ \boldsymbol{g}_2 \\ \vdots \\ \boldsymbol{g}_k \end{pmatrix} = \begin{pmatrix} I_k & A \end{pmatrix},$$

where I_k is the $k \times k$ identity matrix and $A \in \mathbb{F}_2^{k\times(n-k)}$.

Proof **Algorithm construction:**

1. Pick any initial basis of $\mathcal{C}$, arrange its vectors as rows to form a $k \times n$ generator matrix G_0.
2. Perform elementary row operations to transform G_0 into row echelon form, which still shares the same row space $\mathcal{C}$.
3. Permute columns to move a linearly independent $k \times k$ submatrix to the front, then normalize it to the identity matrix I_k.
4. The resulting matrix is $G = \begin{pmatrix} I_k & A \end{pmatrix}$. Take its k rows as $\boldsymbol{g}_1, \boldsymbol{g}_2, \ldots, \boldsymbol{g}_k$, which form a basis of $\mathcal{C}$ and satisfy the required matrix equality.

This completes the construction. ∎

Proposition 7 *Given the generator matrix $G = \begin{pmatrix} I_k & A \end{pmatrix}$ of an $[n,k,d]$ linear code over $\mathbb{F}_2$, one explicit parity-check matrix is*

$$H = \begin{pmatrix} A^\top & I_{n-k} \end{pmatrix}.$$

Proof Compute

$$HG^\top = \begin{pmatrix} A^\top & I_{n-k} \end{pmatrix} \begin{pmatrix} I_k \\ A^\top \end{pmatrix} = A^\top + A^\top = \boldsymbol{0}.$$

This implies $\mathcal{C} \subseteq \ker(H)$.

Clearly $\operatorname{rank}(H) = n-k$, by the rank-nullity theorem

$$\dim\ker(H) = n - \operatorname{rank}(H) = k.$$

Since $\dim \mathcal{C} = k$, we obtain $\mathcal{C} = \ker(H)$.Therefore H is a parity-check matrix. ■

Proposition 8 *Let $\mathcal{C}$ be an $[n, k, d]$-binary code.*

1. *$\mathcal{C}$ can detect up to $d - 1$ bit-flip errors.*
2. *$\mathcal{C}$ can correct up to $\left\lfloor \frac{d-1}{2} \right\rfloor$ bit-flip errors.*

Proof

1. Let $c \in \mathcal{C}$ and $d_H(c, y) = t \le d - 1$. For any $c' \in \mathcal{C}$ with $c' \neq c$, we have $d_H(c, c') \ge d$. By the triangle inequality
$$d_H(y, c') \ge d_H(c, c') - d_H(c, y) \ge d - t \ge 1.$$
Thus $y \notin \mathcal{C}$, so the error is detected.
2. Let $t = \left\lfloor \frac{d-1}{2} \right\rfloor$, so $2t + 1 \le d$. Let $d_H(c, y) \le t$. For any $c' \neq c$,
$$d_H(y, c') \ge d - t > t \ge d_H(c, y).$$
Thus c is the unique nearest codeword, so the error is corrected.

■

Proposition 9 *Consider the ℓ-repetition code.*

1. *The ℓ-repetition code is an $[\ell, 1, \ell]$-linear code over $\mathbb{F}_2$.*
2. *The error-detecting capability is $\ell - 1$.*
3. *The error-correcting capability is $\left\lfloor \frac{\ell-1}{2} \right\rfloor$.*

Proof

1. The encoding function is
$$\mathcal{E} : \mathbb{F}_2 \to \mathbb{F}_2^\ell, \quad \mathcal{E}(0) = (0, \dots, 0), \quad \mathcal{E}(1) = (1, \dots, 1).$$
2. The code has exactly two codewords:
$$\mathcal{C} = \big\{(0, \dots, 0),\ (1, \dots, 1)\big\}.$$
3. The code length is $n = \ell$, the dimension is $k = 1$, and the minimum Hamming distance is
$$d = \min_{c \neq c'} d_H(c, c') = \ell.$$
Thus $\mathcal{C}$ is an $[\ell, 1, \ell]$-linear code.
4. The error-detecting capability is $d - 1 = \ell - 1$. The error-correcting capability is
$$\left\lfloor \frac{d-1}{2} \right\rfloor = \left\lfloor \frac{\ell-1}{2} \right\rfloor.$$

■

Lemma 10 *Let C be a linear code over $\mathbb{F}_2$ with minimum Hamming distance d, and let H be its parity-check matrix. Let $t = \left\lfloor \frac{d-1}{2} \right\rfloor$. Any error pattern $\boldsymbol{e}$ of weight $\mathrm{wt}(\boldsymbol{e}) \le t$ lies in a unique coset of C, and the syndrome $H\boldsymbol{e}^\top$ uniquely determines the coset.*

Proof Suppose for contradiction that there exist two distinct error vectors $\boldsymbol{e}_1, \boldsymbol{e}_2$ with $\mathrm{wt}(\boldsymbol{e}_1), \mathrm{wt}(\boldsymbol{e}_2) \leq t$ such that $\boldsymbol{e}_1 - \boldsymbol{e}_2 \in C$. By the triangle inequality:

$$\mathrm{wt}(\boldsymbol{e}_1 + \boldsymbol{e}_2) \leq \mathrm{wt}(\boldsymbol{e}_1) + \mathrm{wt}(\boldsymbol{e}_2) \leq 2t \leq d - 1.$$

Since $\boldsymbol{e}_1 + \boldsymbol{e}_2 \in C \setminus \{\mathbf{0}\}$, its weight is at least d, a contradiction. Thus, the coset contains at most one error of weight $\leq t$. Moreover, $H\boldsymbol{e}_1^\top = H\boldsymbol{e}_2^\top$ if and only if $\boldsymbol{e}_1 - \boldsymbol{e}_2 \in C$, so syndrome labels the coset uniquely. ■

Theorem 11 *Let $C \subseteq \mathbb{F}_2^n$ be a linear code with minimum distance d, H its parity-check matrix, and let $t = \left\lfloor \frac{d-1}{2} \right\rfloor$. For any transmitted codeword $\mathbf{c} \in C$, error $\mathbf{e}$ with $\mathrm{wt}(\boldsymbol{e}) \leq t$, and received word $\boldsymbol{r} = \boldsymbol{c} + \boldsymbol{e}$, the minimum weight decoding rule uniquely recovers $\mathbf{e}$ and reconstructs the original codeword:*

$$\hat{\boldsymbol{c}} = \boldsymbol{r} + \boldsymbol{e} = \boldsymbol{c}.$$

Proof By the lemma, the coset $\boldsymbol{r} + C$ contains exactly one vector $\boldsymbol{e}_0$ of minimal weight, which must equal the true error $\boldsymbol{e}$. Applying the inverse error yields:

$$\hat{\boldsymbol{c}} = \boldsymbol{r} + \boldsymbol{e}_0 = (\boldsymbol{c} + \boldsymbol{e}) + \boldsymbol{e} = \boldsymbol{c}.$$

Thus, perfect and unique recovery is achieved. ■

$[n, k, d]$-code model:

- $\mathcal{C} \leq \mathbb{F}_2^n, \ \dim \mathcal{C} = k, \ d = \min\limits_{\mathbf{0} \neq \boldsymbol{c} \in \mathcal{C}} \mathrm{wt}(\boldsymbol{c})$
- Message set $\mathcal{M}$, encoding $\mathrm{Enc} : \mathcal{M} \to \mathcal{C}$ bijective.
- $\mathcal{E} = \left\{ \boldsymbol{e} \in \mathbb{F}_2^n \ : \ \mathrm{wt}(\boldsymbol{e}) \leq \left\lfloor \frac{d-1}{2} \right\rfloor \right\}$
- Channel map: $\tau_e(\boldsymbol{c}) := \boldsymbol{c} + \boldsymbol{e}, \ \forall \boldsymbol{c} \in \mathcal{C}$
- Decoding map $\mathrm{Dec} : \mathbb{F}_2^n \to \mathcal{M}$ satisfies $\mathrm{Dec} \circ \tau_e \circ \mathrm{Enc} = \mathrm{Id}_{\mathcal{M}}$.

Algorithm 1 Minimum Weight Decoding (Unique Error Correction)

Require: Parity-check matrix $H \in \mathbb{F}_2^{(n-k)\times n}$, received vector $\boldsymbol{r} \in \mathbb{F}_2^n$, error-correcting capability $t = \left\lfloor \frac{d-1}{2} \right\rfloor$

Ensure: Corrected codeword $\hat{\boldsymbol{c}} \in \mathcal{C}$

1: Compute syndrome: $\boldsymbol{s} \leftarrow H\boldsymbol{r}^\top \in \mathbb{F}_2^{n-k}$
2: Find the unique coset leader $\boldsymbol{e}_0 \in \mathbb{F}_2^n$ satisfying $H\boldsymbol{e}_0^\top = \boldsymbol{s}$ and $\mathrm{wt}(\boldsymbol{e}_0) \leq t$
3: Correct received vector: $\hat{\boldsymbol{c}} \leftarrow \boldsymbol{r} + \boldsymbol{e}_0$
4: **return** $\hat{\boldsymbol{c}}$

5.3 Quantum errors correction

Definition 12 (General Quantum Error Correction Model) A general quantum error correction model consists of four primitive constituents:

1. A finite-dimensional complex physical Hilbert space $\mathcal{H}_{\mathrm{P}}$;
2. A linear code subspace $\mathcal{C} \subseteq \mathcal{H}_{\mathrm{P}}$;
3. A completely positive noise operation $\mathcal{E}\colon \mathcal{L}(\mathcal{H}_{\mathrm{P}}) \to \mathcal{L}(\mathcal{H}_{\mathrm{P}})$;
4. A completely positive recovery operation $\mathcal{R}\colon \mathcal{L}(\mathcal{H}_{\mathrm{P}}) \to \mathcal{L}(\mathcal{H}_{\mathrm{L}})$.

Derived from the above primitive components:

1. The linear operator space $\mathcal{L}(\mathcal{H}_{\mathrm{P}})$ over $\mathcal{H}_{\mathrm{P}}$;
2. The unique orthogonal projection $P \in \mathcal{L}(\mathcal{H}_{\mathrm{P}})$ satisfying $\mathrm{Im}(P) = \mathcal{C}$;
3. The set of code-supported operators

$$\mathcal{S}_{\mathcal{C}} := \left\{ \rho \in \mathcal{L}(\mathcal{H}_{\mathrm{P}}) \,\middle|\, P\rho P = \rho \right\}.$$

The model satisfies the quantum error correction condition if

$$\mathcal{R}\big(\mathcal{E}(\rho)\big) \propto \rho, \quad \forall \rho \in \mathcal{S}_{\mathcal{C}}.$$

i.e. there exists $c > 0$ such that the fo diagram commutes

$$\begin{tikzcd}
\mathcal{S}_{\mathcal{C}} \arrow[rr, "\mathcal{E}"] \arrow[dr, "c\mathrm{Id}"'] & & \mathcal{L}(\mathcal{H}_{\mathrm{P}}) \arrow[dl, "\mathcal{R}"] \\
& \mathcal{S}_{\mathcal{C}} &
\end{tikzcd}$$

Remark 13

$$P\rho P \;:\; \mathcal{H} \xrightarrow{P} \mathcal{H} \xrightarrow{\rho} \mathcal{H} \xrightarrow{P} \mathcal{H};$$

$$P\rho P = \rho \Longleftrightarrow P\rho = \rho P = \rho \Rightarrow \begin{cases} \mathrm{Im}(\rho) \subseteq \mathcal{C} \\ \ker(\rho) \supseteq \mathcal{C}^{\perp} \end{cases}$$

Remark 14 If $\mathcal{E}$ and $\mathcal{R}$ are trace-preserving CPTP channels, the condition reduces to

$$\mathcal{R} \circ \mathcal{E}(\rho) = \rho, \quad \forall \rho \in \mathcal{S}_{\mathcal{C}}.$$

Theorem 15 *Let $\mathcal{E}$ be a quantum operation with Kraus representation*

$$\mathcal{E}(\rho) = \sum_i E_i \rho E_i^{\dagger}.$$

There exists a completely positive recovery operation $\mathcal{R}$ if and only if there exist scalars $\alpha_{ij} \in \mathbb{C}$ such that

$$PE_i^{\dagger} E_j P = \alpha_{ij} P.$$

Proof Please refer to [1] Chap 10. ■

Definition 16 (Stabilizer Code Model) As a subclass of the general quantum error correction model:

1. Equipped with an additional **abelian subgroup** $\mathcal{G}$ of the Pauli group $\mathcal{P}$ on $\mathcal{H}_{\mathrm{P}}$, such that $-\mathbb{I} \notin \mathcal{G}$.
2. The code subspace is defined by
$$\mathcal{C} = \big\{ |\psi\rangle \in \mathcal{H}_{\mathrm{P}} \,\big|\, g|\psi\rangle = |\psi\rangle,\ \forall g \in \mathcal{G} \big\}.$$
3. The noise operation $\mathcal{E}$ is restricted to channels whose Kraus operators belong to $\mathcal{P}$.
4. The recovery operation $\mathcal{R}$ is restricted to coset-based recovery induced by $\mathcal{P}/\mathcal{G}$.

The above structural constraints do not automatically guarantee error recovery;the general error correction condition must still be verified additionally.

Definition 17 Let $\mathcal{P}_n$ be the n-qubit Pauli group. The stabilizer code with $\mathcal{G}$ is denoted by $[[n, k, d]]$, where

1. n: total number of physical qubits.
2. k: dimension of the code subspace (number of logical qubits).
3. For any $P = P_1 \otimes P_2 \otimes \cdots \otimes P_n \in \mathcal{P}_n$ with $P_i \in \{\mathbb{I}, X, Y, Z\}$, define its Pauli weight as
$$\mathrm{wt}(P) \triangleq \# \left\{ i \,\middle|\, 1 \le i \le n,\ P_i \ne \mathbb{I} \right\}.$$
The minimum code distance is
$$d = \min \left\{ \mathrm{wt}(P) \,\middle|\, P \in \mathcal{P}_n,\ P \notin \mathcal{G},\ PG = GP, \forall G \in \mathcal{G} \right\}.$$

Definition 18 (Symplectic Self-Orthogonal Subspace) Let $\mathbb{F}_2^{2n}$ be the $2n$-dimensional vector space over the binary field $\mathbb{F}_2$, equipped with the **symplectic inner product**
$$\boldsymbol{u} \odot \boldsymbol{v} = \boldsymbol{a}_1 \cdot \boldsymbol{b}_2 + \boldsymbol{a}_2 \cdot \boldsymbol{b}_1 \in \mathbb{F}_2,$$
where $\boldsymbol{u} = (\boldsymbol{a}_1, \boldsymbol{b}_1), \boldsymbol{v} = (\boldsymbol{a}_2, \boldsymbol{b}_2) \in \mathbb{F}_2^{2n}$ and $\cdot$ denotes the standard dot product in $\mathbb{F}_2^n$.
A linear subspace $C \subseteq \mathbb{F}_2^{2n}$ is called **symplectic self-orthogonal** if
$$\boldsymbol{u} \odot \boldsymbol{v} = 0 \quad \text{for all } \boldsymbol{u}, \boldsymbol{v} \in C.$$

Theorem 19 *Given a positive integer n, the composition map*
$$\Psi : \mathcal{P}_n \xrightarrow{\pi} \mathcal{P}_n/\langle iI \rangle \xrightarrow{\Phi} \mathbb{F}_2^{2n}$$
induces a surjective map
$$\big\{ G \le \mathcal{P}_n \,\big|\, G \textit{ is abelian and } -I \notin G \big\} \xrightarrow{\widetilde{\Psi}} \big\{ C \le \mathbb{F}_2^{2n} \,\big|\, C \textit{ is symplectic self-orthogonal} \big\}.$$
Furthermore, if $C = \mathbb{F}_2 c_1 \oplus \cdots \oplus \mathbb{F}_2 c_k$,then there exist $g_j \in \Psi^{-1}(c_j)$ for each j, such that
$$G = \langle g_1, \ldots, g_k \rangle$$
and $\widetilde{\Psi}(G) = C$.

Proof Let
$$\mathcal{S} = \big\{ G \le \mathcal{P}_n \mid G \text{ abelian},\ -I \notin G \big\}, \quad \mathcal{C} = \big\{ C \le \mathbb{F}_2^{2n} \mid C \text{ symplectic self-orthogonal} \big\}.$$

Define $\widetilde{\Psi}(G) = \Psi(G)$. Since Ψ is a group homomorphism from $\mathcal{P}_n$ to $\mathbb{F}_2^{2n}$, $\Psi(G)$ is a linear subspace of $\mathbb{F}_2^{2n}$. For any $g_1, g_2 \in G$, G being abelian gives $g_1 g_2 = g_2 g_1$, which is equivalent to

$$\langle \Psi(g_1), \Psi(g_2) \rangle_s = 0.$$

Thus $\Psi(G)$ is symplectic self-orthogonal, so $\widetilde{\Psi} : \mathcal{S} \to \mathcal{C}$ is well-defined.

Next prove surjectivity. Take any $C \in \mathcal{C}$, choose a basis $c_1, \dots, c_k$ of C, so

$$C = \mathbb{F}_2 c_1 \oplus \cdots \oplus \mathbb{F}_2 c_k.$$

For each j, the preimage $\Psi^{-1}(c_j)$ is a coset of $\langle iI \rangle$, hence of the form $\{P_j, iP_j, -P_j, -iP_j\}$. If $P_j^2 = I$, set $g_j = P_j$; if $P_j^2 = -I$, set $g_j = iP_j$. Then

$$g_j^2 = (iP_j)^2 = i^2 P_j^2 = -(-I) = I,$$

so in either case we obtain $g_j \in \Psi^{-1}(c_j)$ with $g_j^2 = I$.

Let $G = \langle g_1, \dots, g_k \rangle$. Since $\langle c_i, c_j \rangle_s = 0$ for all i, j, each pair g_i, g_j commutes, so G is abelian. Suppose $-I \in G$, then there exist $a_j \in \mathbb{F}_2$ such that

$$-I = \prod_{j=1}^{k} g_j^{a_j}.$$

Applying Ψ,

$$0 = \Psi(-I) = \sum_{j=1}^{k} a_j c_j.$$

By linear independence of $\{c_j\}$, all $a_j = 0$, which implies $-I = I$, a contradiction. Hence $-I \notin G$, so $G \in \mathcal{S}$.

By construction, $\widetilde{\Psi}(G) = \operatorname{span}\{c_1, \dots, c_k\} = C$. Therefore $\widetilde{\Psi}$ is surjective, and the existence of such g_j and G follows immediately. ■

Corollary 20 *Let $C \leq \mathbb{F}_2^{2n}$ be any symplectic self-orthogonal subspace, rhen*

$$\left|\widetilde{\Psi}^{-1}(C)\right| = 2^{\dim C}.$$

Proof Choose a basis $\{c_1, \dots, c_k\}$ of C. For each c_j, pick a fixed preimage $g_j \in \Psi^{-1}(c_j)$ with $g_j^2 = I$. Any element $G \in \widetilde{\Psi}^{-1}(C)$ can be generated by $\varepsilon_j g_j$ with $\varepsilon_j \in \{+1, -1\}$ for each j. Each choice of $(\varepsilon_1, \dots, \varepsilon_k)$ gives a distinct abelian subgroup G satisfying $-I \notin G$, and all such G are obtained in this way. There are exactly $2^k = 2^{\dim C}$ independent sign combinations, hence $\left|\widetilde{\Psi}^{-1}(C)\right| = 2^{\dim C}$. ■

Definition 21 (The **CSS code** over $\mathbb{F}_2$)

1. Let $\mathcal{C}_1, \mathcal{C}_2 \subseteq \mathbb{F}_2^n$ be two binary linear codes satisfying $\mathcal{C}_1 \perp \mathcal{C}_2$, i.e. $\mathcal{C}_1 \cdot \mathcal{C}_2 = \{0\}$.
2. Let $\dim \mathcal{C}_1 > \dim \mathcal{C}_2$.

The **CSS code** over $\mathbb{F}_2$ constructed from $\mathcal{C}_1^\perp \oplus \mathcal{C}_2$ is an $[n, k, d]$ linear code with

$$k = \dim \mathcal{C}_1^\perp + \dim \mathcal{C}_2, \quad d = \min\left\{d(\mathcal{C}_1^\perp),\, d(\mathcal{C}_2)\right\}.$$

Remark 22 The **CSS code** over $\mathbb{F}_2$ is a symplectic self-orthogonal subspace:

For $\boldsymbol{u} = (\boldsymbol{a}_1, \boldsymbol{b}_1), \boldsymbol{v} = (\boldsymbol{a}_2, \boldsymbol{b}_2) \in \mathcal{C}$, we know $\mathcal{C}_1 \cdot \mathcal{C}_2 = \{0\} \implies \boldsymbol{a}_1 \cdot \boldsymbol{b}_2 = \boldsymbol{a}_2 \cdot \boldsymbol{b}_1 = 0$ for all $\boldsymbol{a}_1, \boldsymbol{a}_2 \in \mathcal{C}_1, \boldsymbol{b}_1, \boldsymbol{b}_1 \in \mathcal{C}_2$. Thus

$$\boldsymbol{u} \odot \boldsymbol{v} = \boldsymbol{a}_1 \cdot \boldsymbol{b}_2 + \boldsymbol{a}_2 \cdot \boldsymbol{b}_1 = 0.$$

Example 23 (Three-qubit bit-flip code) This is a stabilizer code with generators

$$S_1 = Z_1 Z_2, \quad S_2 = Z_2 Z_3.$$

It detects and corrects single X-errors, but cannot detect Z-errors. Parameters: $[[3, 1, 1]]$.

Example 24 (Shor code) A $[[9, 1, 3]]$ stabilizer code formed by concatenating bit-flip and phase-flip repetition codes. It corrects arbitrary single-qubit X, Y, Z-errors using Z-type and X-type stabilizers.

Example 25 (Steane code) A $[[7, 1, 3]]$ CSS stabilizer code built from the classical $[7, 4, 3]$ Hamming code. Stabilizers are of the form $X_1 X_2 X_3 X_4$ and $Z_1 Z_2 Z_3 Z_4$, mutually commuting. It corrects any single-qubit error.

Example 26 (Steane Code) Take classical $[7, 4, 3]$ Hamming code $\mathcal{C}$. Set $\mathcal{C}_1 = \mathcal{C}$, $\mathcal{C}_2 = \mathcal{C}^{\perp}$, then $\mathcal{C}_2 \subseteq \mathcal{C}_1$. This yields the $[[7, 1, 3]]$ Steane CSS code, which is a stabilizer code correcting all single-qubit errors.

5.4 Plan for next week

1. TextBook chap 10.

5.5 Appendix

Lemma 27 *Let $\mathcal{H}_1, \mathcal{H}_2$ be finite-dimensional complex Hilbert spaces, and let $A\colon \mathcal{H}_1 \to \mathcal{H}_2$ be injective linear. Define $\mathbb{P}\mathcal{H} := \big(\mathcal{H} \setminus \{0\}\big)/\mathbb{C}^*$. Then A induces a well-defined map $\widetilde{A}\colon \mathbb{P}\mathcal{H}_1 \to \mathcal{L}(\mathcal{H}_2)$ by*

$$\widetilde{A}\big([\psi]\big) := \frac{A|\psi\rangle\big(A|\psi\rangle\big)^\dagger}{\langle \psi \mid \psi\rangle}.$$

Proof Let $[\psi] = [\phi]$, so $\psi = \lambda\phi$ for some $\lambda \in \mathbb{C}^*$. By linearity, $A\psi = \lambda A\phi$, and

$$A\psi(A\psi)^\dagger = |\lambda|^2\, A\phi(A\phi)^\dagger, \quad \langle \psi \mid \psi\rangle = |\lambda|^2\, \langle \phi \mid \phi\rangle.$$

Cancelling the positive factor $|\lambda|^2$ yields

$$\frac{A\psi(A\psi)^\dagger}{\langle \psi \mid \psi\rangle} = \frac{A\phi(A\phi)^\dagger}{\langle \phi \mid \phi\rangle}.$$

The value is independent of the choice of representative, hence $\widetilde{A}$ is well-defined. ■

Definition 28 The quantum error-correction model consists of

- $\mathcal{H}_{\mathrm{L}}, \mathcal{H}_{\mathrm{P}}$: finite-dimensional complex Hilbert spaces;
- $\mathbb{P}(\mathcal{H}_{\mathrm{L}})$: projective space of rays in $\mathcal{H}_{\mathrm{L}}$;
- $\mathcal{L}(\mathcal{H}_{\mathrm{P}})$: space of linear operators on $\mathcal{H}_{\mathrm{P}}$;
- $\mathcal{E}\colon \mathcal{L}(\mathcal{H}_{\mathrm{P}}) \to \mathcal{L}(\mathcal{H}_{\mathrm{P}})$: quantum channel;
- $\mathrm{Enc}\colon \mathcal{H}_{\mathrm{L}} \to \mathcal{H}_{\mathrm{P}}$: encoding map;
- $\mathrm{Dec}\colon \mathcal{L}(\mathcal{H}_{\mathrm{P}}) \to \mathbb{P}(\mathcal{H}_{\mathrm{L}})$: decoding map.

The model is error-correcting if

$$\mathrm{Dec} \circ \mathcal{E} \circ \widetilde{\mathrm{Enc}} = \mathrm{Id}_{\mathbb{P}(\mathcal{H}_{\mathrm{L}})}.$$

Equivently, the diagram commutes

$$\begin{tikzcd}
\mathbb{P}\mathcal{H}_{\mathrm{L}} \arrow[r, "\widetilde{\mathrm{Enc}}"] \arrow[d, "\mathrm{Id}"'] & \mathcal{L}(\mathcal{H}_{\mathrm{P}}) \arrow[d, "\mathcal{E}"] \\
\mathbb{P}\mathcal{H}_{\mathrm{L}} & \mathcal{L}(\mathcal{H}_{\mathrm{P}}) \arrow[l, "\mathrm{Dec}"']
\end{tikzcd}$$

$$\mathbb{P}\mathcal{H}_L \xrightarrow{\widetilde{\mathrm{Enc}}} \mathcal{L}(\mathcal{H}_P) \xrightarrow{\mathcal{E}} \mathcal{L}(\mathcal{H}_P) \xrightarrow{\mathcal{R}} \mathbb{P}\mathcal{H}_L$$

Lemma 29 *Denote $\mathcal{Q} = \operatorname{Im}\mathrm{Enc}$ the quantum code subset of $\mathcal{L}(\mathcal{H}_P)$, define the recovery map $\mathcal{R} := \widetilde{\mathrm{Enc}} \circ \mathrm{Dec}$. Then we have*

$$\mathcal{R} \circ \mathcal{E} = \mathrm{Id}_{\mathcal{Q}}.$$

The quantum error-correction model can be simplified as

$$\begin{tikzcd}
\mathcal{Q} \arrow[rr, equal, "\mathrm{Id}"] \arrow[dr, "\mathcal{E}"'] & & \mathcal{Q} \\
& \mathcal{L}(\mathcal{H}_{\mathrm{P}}) \arrow[ur, "\mathcal{R}"'] &
\end{tikzcd}$$

References

[1] M. A. Nielsen and I. L. Chuang. *Quantum Computation and Quantum Information*, 10th Anniversary Edition. Cambridge University Press, 2010.

[2] A. Pesah. Classical error correction. *Personal Blog*, May 2022. URL: https://arthurpesah.me/blog/2022-05-21-classical-error-correction/